

TEACHER MASTER KEY & GRADING RUBRIC

Constant Speed Story: Dual-Graph Edition • Day 15 Performance Task

TEACHER REFERENCE KEY

10-POINT HOLISTIC GRADING RUBRIC:

1. Illustration (2 pts): Clear moving object, origin (0,0) labeled, motion direction vector arrow.

2. Word Problem (2 pts): Authentic constant-speed story stating 2 knowns with units and 1 clear target unknown.

3. GUESS Proof (3 pts): G, U, E, S, S completely filled out, algebra correct, final answer boxed with units.

4. Dual Graphs (3 pts): Calibrated axes, straight line on x-t (slope = v), horizontal line on v-t (area = d).

☀ COMPLETE STUDENT WORK EXEMPLAR: "THE MARS ROVER TRAVERSE"

Student Story / Problem:

"NASA's autonomous Mars Recon Rover drives across the flat plains of Jezero Crater at a steady speed of **15.0 m/s** for a total time of **40.0 seconds**. How many meters of Martian terrain does the rover cross before its next sensor scan?"

Complete GUESS Mathematical Proof:

G Speed $v = 15.0 \text{ m/s}$ • Time $t = 40.0 \text{ s}$

U Distance $d = ? \text{ meters}$

E $d = v \cdot t$

S $d = (15.0 \text{ m/s}) \cdot (40.0 \text{ s})$

S **$d = 600 \text{ meters}$ [Boxed]**

Scale Planning Choices:

- Time (Horizontal):** 10 blocks = 50 s (5 s per block). Point plotted at $t = 40 \text{ s}$ (8th line).
- Position x (Graph 1):** 8 blocks = 800 m (100 m per block). Plotted from (0, 0) to (40, 600).
- Velocity v (Graph 2):** 6 blocks = 30 m/s (5 m/s per block). Horizontal line at $v = 15 \text{ m/s}$ (3rd line).

Synthesis Answers:

- Slope = **velocity / speed** ($600\text{m} / 40\text{s} = 15 \text{ m/s}$).
- Area = **distance / displacement** ($15 \text{ m/s} \times 40 \text{ s} = 600 \text{ m}$).
- If 2× speed: Slope doubles (steeper line); Area doubles (box is twice as tall = 1,200 m).

Graph 1: x vs. t Visual Check

- Line **MUST** be straight (not curved).
- Starts at origin (0, 0) and ends at (40 s, 600 m).
- Slope = Rise / Run = $600 / 40 = 15.0 \text{ m/s}$.

Graph 2: v vs. t Visual Check

- Line **MUST** be perfectly horizontal at height $v = 15 \text{ m/s}$.
- Extends from $t = 0$ to $t = 40 \text{ s}$.
- Shaded Rectangle Area = $15 \text{ m/s} \times 40 \text{ s} = 600 \text{ m}$.

Common Student Pitfalls to Watch For:

- Curved Lines:** Students drawing curves instead of straight lines (constant speed = linear x-t, horizontal v-t).
- Unit Dropping:** Forgetting units on Givens or final answer (e.g. writing "600" instead of "600 m").
- Slanted v-t line:** Students drawing a sloped line on v-t (remind: sloped v-t means accelerating!).
- Misaligned Time Scale:** Using different time scales on Graph 1 and Graph 2. They should align!